Trustworthy Explanations for Student Dropout Risk: Comparing Black-Box and Interpretable Models

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1. Introduction, Aim and Research Questions

Early identification of dropout risk can enable timely academic or financial support, but an unreliable explanation can misdirect scarce advising resources or produce unjustified conclusions about a student. This project will examine first-year undergraduate dropout risk using structured data from the UCI Predict Students' Dropout and Academic Success dataset and TabICL, a peer-reviewed tabular foundation model (TFM) for in-context learning (Qu et al., 2025). TFMs achieve competitive results on small tabular datasets (Hollmann et al., 2025), but explanation quality is multidimensional and must be evaluated quantitatively and with users, not through anecdotal plots alone (Nauta et al., 2023). The principal contribution will therefore be evidence about the reliability and practical value of local explanations. Three questions will guide the study: RQ1: does TabICL add predictive value over simpler baselines; RQ2: are its local explanations faithful and stable; RQ3: do they help users anticipate model outputs and identify feasible support actions? Real-data model results, explanation audits and human-study findings will be reported separately, and explanations will be presented as descriptions of model associations, not as causes of dropout.

All predictive modelling and explanation evaluation will use one openly licensed UCI dataset. Short, consented written responses from the human evaluation will form the only unstructured component and will remain linked to the study cases, keeping the prediction task, explanations and human evidence coherently connected.

2. Decision Context and Intended User

The intended decision context is a university student-support officer considering currently enrolled first-year students immediately after first-semester results are finalised and before second-semester support decisions; this project will simulate the described context using historical, pseudonymised UCI cases. For a student flagged as high risk, the officer would decide whether to prioritise that student for an outreach meeting and referral to appropriate academic or financial support. Missing a genuinely at-risk student may prevent timely contact, whereas excessive flagging consumes limited support capacity and can erode trust. The explanation must therefore clarify the model's output without presenting associations as causal diagnoses or recommendations.

3. Data Sources

Primary modelling data. 'Predict Students' Dropout and Academic Success' (Realinho, Vieira Martins, Machado, & Baptista, 2021), UCI Machine Learning Repository dataset 697, is distributed under CC BY 4.0 and documented in a peer-reviewed data paper (Realinho et al., 2022). It contains 4,424 student records and 36 predictors spanning demographic, socio-economic, admission, administrative, academic and macroeconomic information, with labels Dropout, Enrolled and Graduate. The distributed version has no direct identifiers, and exploratory analysis found no missing values or duplicate rows.

Prospective qualitative data. A voluntary convenience sample from the Asian Institute of Technology (AIT) community will provide brief written rationales, suggested support actions and comments on missing information while evaluating pseudonymised UCI cases. These consented, pseudonymised responses will form the project's unstructured dataset, will be linked to pseudonymous participant and case IDs, and will be analysed qualitatively for comprehension, actionability and information gaps. No AIT student records will be collected, and responses will not enter the dropout model or be presented as explanations of real outcomes.

4. Prediction Task and Study Design

The primary prediction task will be binary: Dropout (1) versus Graduate (0). Enrolled records will be excluded as unresolved outcomes rather than confirmed non-dropout cases; this will simplify interpretation but may introduce selection bias and limit applicability to similar unresolved students. A meaningful share of Dropout records (17.3%, versus 3.4% of Graduate records) show no first-semester evaluations, so some labelled dropouts may have left before the decision point rather than after it; this timing ambiguity will be reported as an explicit eligibility limitation. The resulting class distribution will be documented. Subject to time and class support, sensitivity analysis will retain Enrolled as a separate class or group it with non-dropout, while primary conclusions will stay tied to the pre-specified binary task.

The prediction time will be immediately after first-semester results. Only variables available by then will enter the model: enrolment/admission records, demographic and socio-economic fields, administrative status, macroeconomic indicators, and first-semester outcomes. Second-semester curricular variables will be excluded as temporal leakage. First-semester grades and approved units are predictive but already historical, so they will be treated as immutable in counterfactual recourse.

5. Proposed Methodology

5.1 Predictive Modelling and Evaluation

A fixed, stratified 70/15/15 train/validation/test split supports identical comparisons. Before test access, protocol amendments added a native main-effects Explainable Boosting Machine (EBM) and froze the audit rules without changing the split, seed, target, feature windows, capacity policy or evaluation rules. Validation ROC-AUC was 0.952 for TabICL 2.2.0 (four estimators), 0.946 for EBM, 0.945 for logistic regression and XGBoost, and 0.923 for a depth-3 tree. These validation-only results do not establish TabICL's superiority; its 0.006 validation ROC-AUC margin over EBM is modest. EBM uses raw named variables, explicit nominal types and no interactions; its additive-model rationale follows Lou et al. (2013). The policy ranks each cohort and flags the top 20% with stable tie-breaking; no universal probability threshold is claimed. At the validation top-20% capacity, TabICL flagged 108 of 544 cases, achieving the capacity-limited maximum recall of 108/213 (0.507) with 1.000 precision. A separate admission-time validation comparison, using only enrolment/admission features and excluding first-semester outcomes, gave ROC-AUCs of 0.8753 for EBM, 0.8752 for XGBoost and 0.8739 for TabICL; these are not the primary post-first-semester task results. After all model and explanation choices are frozen, one final same-test-set evaluation will report discrimination, capacity metrics, calibration, runtime and paired 2,000-resample bootstrap intervals for model differences. The pre-specified top-20% capacity will remain the primary operating point; any alternative capacities will be supplementary.

5.2 Explanation Method and Faithfulness Evaluation

As a TFM is not a tree ensemble, TreeSHAP exactness will not be claimed. Model-agnostic permutation SHAP (Lundberg & Lee, 2017) will explain dropout probability using 25 training-only background rows and a max_evals budget of 67 model evaluations per explained case; 67 is the explainer budget, not a count of cases. The local faithfulness and stability audits will use the same 20 cases described in Section 5.3, selected from the validation split by predicted-risk stratum using predictions only, so neither test data nor outcome labels influence case selection or tuning. Signed member attributions will be summed within each semantic group, preserving additivity, before groups are ranked by absolute total. Deletion will mask those same groups with training references, recompute derived variables, and compare the cumulative absolute probability change with 50 random group orders; the primary statistic will be ranked-minus-random area with case-level uncertainty. Similar curves will indicate inadequate fidelity, and masking will measure model reliance, not intervention, and attributions describe model associations, not causes of dropout. Validation checks reconstructed TabICL probabilities within 4.1e-8 and recovered known white-box attribution ranks with Spearman 1.0. Grouped permutation importance will supply global reliance, while ALE will examine three validation-selected numerical effects with less off-manifold risk than PDP under correlation.

5.3 Explanation Stability Evaluation

The stability and local faithfulness evaluations will use the same 20 validation cases, selected deterministically by taking four cases from each of five predicted-risk strata; selection uses predictions only and never outcome labels, and the test split stays unopened. For each case, the explainer will run 10 times, varying seed and training-background sample separately. Mean pairwise top-five overlap—shared features divided by five—will be primary, with rank correlation secondary. Continuous perturbations will draw Gaussian noise with standard deviation equal to 1% of the feature's training standard deviation, clip to training bounds, recompute dependencies, and qualify only when absolute probability change is at most 0.01. Mean overlap below the validation-frozen 0.80 warning level will be displayed, not hidden through selective reruns (Tiukhova et al., 2024).

5.4 Human Evaluation

Following forward-simulation evaluation (Hase & Bansal, 2020), the human study will aim to recruit 8–10 voluntary AIT participants. At least four participants will be required, and the achieved sample will be allocated as evenly as practicable between groups using seeded assignment. After standardised instructions and one unscored practice case, both groups will receive the same label-free panel of five flagged and five unflagged profiles in random order; the explanation group will additionally see feature contributions. Neither group will receive the output, probability, capacity boundary or correctness feedback until all 10 responses are locked. The interface will record predictions, confidence, time and written responses. Participant-level accuracy will be primary; group accuracy and time will be compared descriptively alongside illustrative paraphrased comments and coded responses (Section 5.7). Recruitment shortfalls and unequal group sizes will be reported transparently. Unless support professionals participate, the study will test proxy-user comprehensibility, not professional effectiveness, and will make no significance claim. Optional extension: if at least 12 participants are recruited, the explanation group will be split into a contribution chart only and a chart plus a plain-language narrative rendered by a verified local template from the same contributions, with no free-form generated text; with fewer participants this condition is omitted. Any such comparison is exploratory and reported separately from the model results.

5.5 Actionability and Counterfactual Analysis

Before modelling, features will be classified by temporal availability and feasibility. Age, nationality, prior qualifications and completed first-semester outcomes will remain fixed. Only debtor and tuition-fee status will be enumerated as potentially resolvable administrative conditions, and only when the resolved value is valid and realistic before second-semester support decisions; any change may require institutional support and will not be framed as a student-controlled obligation. For a flagged case, the search will find the smallest feasible change that moves it outside the cohort's top-20% set after recomputing its rank while all other scores stay fixed. No universal probability threshold will be assumed, and cases with no feasible recourse will be reported explicitly. These outputs describe model behaviour, not causal promises or guaranteed outcomes (Karimi et al., 2021).

5.6 System Architecture, Workflow and Expected Outputs

The end-to-end workflow will deliver a frozen comparison of TabICL and baselines, explanation and recourse evidence, subgroup diagnostics, human-study findings, a model card and reproducible reports. A lightweight browser-based research interface will provide decision support with frozen predictions, local explanations, reliability warnings and feasible support options, plus a blinded-study mode. The blinded study will withhold model outputs and feedback until all 10 scored responses are submitted. The interface will render versioned, precomputed Python outputs (predictions, contributions and reliability warnings) rather than run live inference, run locally or in a controlled course environment, and accept only pre-approved, pseudonymised UCI test cases. Case packs and response submissions will be validated, rejecting missing, extra, invalid or out-of-range fields rather than silently changing them. Model weights, raw participant responses and credentials will never enter frontend code or version control, and an audit panel will display model, data-split and interface versions for traceability. Before test access, interface parity is checked with training/validation fixtures; study cases are loaded only after the single final evaluation, followed by an export-parity check without retuning. It is a research demonstration, not a production AIT service.

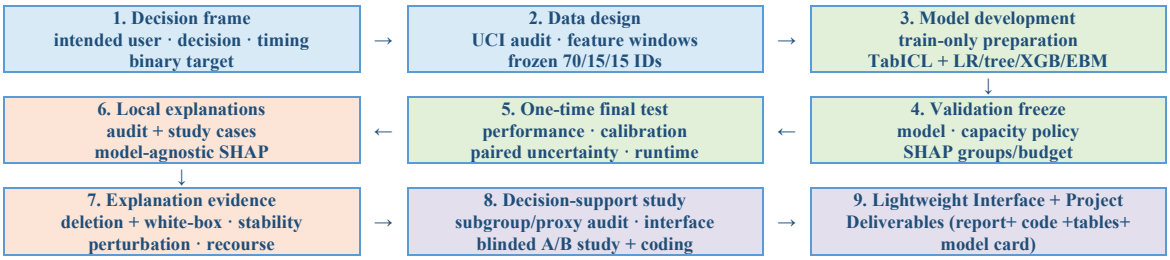


Figure 1. Proposed research workflow. Model, capacity policy and explanation protocols will be frozen on validation data before one-time final testing. Ethics, governance and reproducibility will apply throughout.

5.7 Qualitative Analysis of Human-Study Responses

After the prediction task, each participant will explain the judgement, propose a support action, and flag missing information. A pre-defined coding guide will group responses into comprehension, actionability and information-gap categories. Action suggestions will additionally be coded for whether they identify a concrete next step, are feasible before second-semester decisions, and distinguish student-controlled from institution-supported actions; unsupported causal assumptions will be flagged. Two reviewers will independently code at least 20% of responses to report raw agreement, then resolve disagreements and report aggregate counts with paraphrased examples. This qualitative analysis will complement the forward-simulation results but will not train, validate or explain the dropout model.

6. Ethics, Fairness and Data Governance

The UCI dataset is publicly available under CC BY 4.0 with no direct identifiers; its citation and licence will be preserved, the raw file will remain unchanged, and all transformations will be documented. Human-study participants will be consenting adults (18 or older) recruited voluntarily from the AIT community; participation or refusal will have no effect on grades, employment or access to services. Consent information will explain the task, recorded data, withdrawal limit and retention plan before participation. No names or institutional identifiers will be collected in study responses; random participant IDs and pseudonymised case IDs will be used, and any case lookup will be stored separately with access limited to the research team. Raw free text will stay outside version control and accidental identifiers will be removed; pseudonymous does not mean fully anonymous, so raw-response access will be restricted to the research team. Participants may request withdrawal using their participant ID until their response has been de-identified and included in aggregate analysis; this limit will be stated during consent. As a class project, the study follows basic research-ethics practice (voluntary informed consent, minimal data, no identifiers, restricted access and a stated retention limit) rather than a formal institutional ethics-board process, and collection will stay within the consent text shown to participants. Before recruitment, the team will also record the applicable course retention deadline; raw responses and case-linkage files will be securely deleted at that deadline, retaining only non-identifiable aggregate results. The interface will run locally or in a controlled course environment. Public deployment is outside the scope of this project and would require separate privacy, security and ethics review.

Protected attributes—gender, nationality, age, international status, displaced status and special needs—will be treated as explicit audit attributes. Every group count will be retained; performance metrics will be suppressed rather than silently omitted below 30 records, and unavailable comparisons will be stated. Eligible groups will receive flag rate, true- and false-positive rates, precision, mean risk and calibration gap under the exact capacity labels. All non-protected model features will be screened as potential proxies using Cramér's V for categorical features and correlation-ratio eta for continuous features. These diagnostics will be descriptive, not proof of fairness.

Plausible proxies will also be examined: parental occupation and qualification may encode socio-economic position, course and application mode may reflect demographic sorting, and debtor, tuition-fee and scholarship variables may encode financial disadvantage. Removing a protected attribute does not eliminate correlated proxy information, so influential proxies and limitations will be documented even where they weaken the model's apparent suitability. The system will support outreach prioritisation, not autonomous denial of education or services, and it must not frame financial disadvantage as a personal obligation.

7. Feasibility and Model Complexity

TabICL combines learned representations with in-context information from a reference set and fails simulatability and decomposability: its reasoning cannot be recovered from a coefficient list or small tree, motivating Section 5.2's post-hoc protocol. By contrast, the main-effects EBM is intrinsically interpretable: each prediction is exactly the logistic transform of an intercept plus one learned shape-function contribution per feature. Its validation probabilities reconstruct from those terms within 3.3e-16, and global term strengths plus fixed low-, median- and high-risk local decompositions are exported. Logistic regression and the depth-3 tree provide further simulatable comparisons. TabICL's validation ROC-AUC exceeds EBM by only 0.006, so the project does not presume that complexity is operationally justified; final same-test-set performance, calibration, runtime and explanation evidence will determine whether TabICL remains a research comparison or EBM is the more defensible deployment candidate.

8. Reproducibility and Documentation

The repository separates source data, EDA, modelling, explanation evaluation, study materials and outputs. A pinned environment supports a fresh clone. Seed 42 and source-row IDs define the split; protocol 1.3, 26 in-scope tests, four baselines, capacity-consistent recourse, exact EBM decomposition, TabICL/SHAP integration, grouped importance, ALE, explicit subgroup suppression and proxy tables are implemented. Before final submission, one command will regenerate every report artefact and another will launch the interface from frozen artefacts. The completed model card will record intended use, data, final metrics, explanation evidence, subgroup results and limitations.

9. Expected Contribution, Limitations and Contingencies

The proposed contribution will extend beyond accuracy by asking whether TabICL's post-hoc local explanations merit trust and whether an intrinsically interpretable EBM offers a more defensible alternative. Same-test-set baselines, deletion faithfulness, white-box recovery, repeated-run and perturbation stability, human forward simulation, qualitative evidence and feasibility-constrained counterfactuals will form a coherent evaluation. Expected limitations include the single-institution UCI context, exclusion of unresolved Enrolled outcomes, approximate model-agnostic attribution, non-causal observational data, a small convenience sample, and likely non-professional audience proxies; therefore, the AIT study will test comprehensibility, not predictive validity or deployment suitability at AIT. If recruitment reaches only the minimum sample, the study will remain descriptive. Findings will remain valuable if EBM matches or exceeds TabICL, explanations are unstable or unfaithful, participants gain no benefit, or feasible recourse is unavailable. Explanations describe model associations, not causal diagnoses or guaranteed intervention benefits, and real-data predictive evidence, explanation audits and human-study findings will not be merged into a single claim.

Code and data availability. The versioned project repository is available at https://github.com/TheInterpreters/Student-Dropout-Risk

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